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### Fence with integrated guard

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#### Abstract

A fence with an integrated guard includes a fence having a bottom end and a guard attached to the lower end of the fence. The guard can wrap around the bottom end of the fence. Alternatively, the guard can extend downwardly from the fence. Still alternatively, the guard can extend between adjacent horizontal members of the fence.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims priority from U.S. Provisional Patent Application Ser. No. 62/554,599, filed on Sep. 6, 2017, which is incorporated herein by reference in its entirety.

### BACKGROUND OF THE INVENTION

#### Field of the Invention

(1) The present invention relates to wire fencing and, more specifically, to a wire fence having an integrated guard that is used to line the lower perimeter of the fence, preventing the growth of grass and weeds therein, as well as providing a constant barrier against which a rotating weed cutting device can be applied to reduce the breakage of weed cutting elements and reduce damage to the wire fence.

#### Description of the Related Art

(2) Those who use metal mesh fencing to enclose an outdoor space know all too well of the

difficulties associated with keeping weeds and grasses from growing in between the structural members of the fence. The grass and weeds get entangled in the fence mesh, making it difficult to remove, even with conventional powered rotating line cutters. The metal mesh tends to shield the grass and weeds and also tends to sever the trimmer filament line.

(3) It would be beneficial to provide a fence guard that prevents or growth of grass and weeds in the mesh of a metal mesh fence, and also provides a surface that enables the use of rotating line cutters that will reduce the breakage of the filament lines used therewith.

#### SUMMARY OF THE INVENTION

(4) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

(5) In one embodiment, the present invention is a mesh fence assembly including a mesh fence and a lower border guard that prevents the growth of weeds and grass in between mesh members.

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## Description

#### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated herein and constitute part of this specification, illustrate the presently preferred embodiments of the invention, and, together with the general description given above and the detailed description given below, serve to explain the features of the invention. In the drawings:

(2) FIG. 1 is a perspective view of a fence assembly according to an exemplary embodiment of the present invention;

(3) FIG. 2 is a sectional view of the fence assembly of FIG. 1, taken along lines 2-2 of FIG. 1;

(4) FIG. 3 is a side elevational view of a fence assembly according to an alternative embodiment of the present invention;

(5) FIG. 4 is a side elevational view of a fence assembly according to another alternative embodiment of the present invention;

(6) FIG. 5 is a perspective view of a grass mat according to an exemplary embodiment of the present invention;

(7) FIG. 6 is a perspective view of a grass mat according to an alternative exemplary embodiment of the present invention; and

(8) FIG. 7 is a side elevational view of the grass mat of FIG. 6.

#### DETAILED DESCRIPTION

(9) In the drawings, like numerals indicate like elements throughout. Certain terminology is used herein for convenience only and is not to be taken as a limitation on the present invention. The terminology includes the words specifically mentioned, derivatives thereof and words of similar import. The embodiments illustrated below are not intended to be exhaustive or to limit the invention to the precise form disclosed. These embodiments are chosen and described to best explain the principle of the invention and its application and practical use and to enable others skilled in the art to best utilize the invention.

(10) Reference herein to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment can be included in at least one embodiment of the invention. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments necessarily mutually exclusive of other embodiments. The same applies to the term “implementation.”

(11) As used in this application, the word “exemplary” is used herein to mean serving as an

example, instance, or illustration. Any aspect or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs. Rather, use of the word exemplary is intended to present concepts in a concrete fashion.

(12) Additionally, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or”. That is, unless specified otherwise, or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

(13) Unless explicitly stated otherwise, each numerical value and range should be interpreted as being approximate as if the word “about” or “approximately” preceded the value of the value or range.

(14) The use of figure numbers and/or figure reference labels in the claims is intended to identify one or more possible embodiments of the claimed subject matter in order to facilitate the interpretation of the claims. Such use is not to be construed as necessarily limiting the scope of those claims to the embodiments shown in the corresponding figures.

(15) It should be understood that the steps of the exemplary methods set forth herein are not necessarily required to be performed in the order described, and the order of the steps of such methods should be understood to be merely exemplary. Likewise, additional steps may be included in such methods, and certain steps may be omitted or combined, in methods consistent with various embodiments of the present invention.

(16) Although the elements in the following method claims, if any, are recited in a particular sequence with corresponding labeling, unless the claim recitations otherwise imply a particular sequence for implementing some or all of those elements, those elements are not necessarily intended to be limited to being implemented in that particular sequence.

(17) The present invention provides a temporary appliqué to the front of a cricket bat that provides a visual indication of where on the bat that a cricket ball struck during a swing. While one use of the present invention is, those skilled in the art will recognize that the present device can have other applications as well.

(18) Referring to the Figures, a mesh fence assembly **100** (“assembly **100**”) according to an exemplary embodiment of the present invention is shown. Assembly **100** includes a wire mesh fence **110** and an integrated guard **140**. Fence **100** has a top end **112** and a bottom end **114**, with a first plurality of horizontal members **120** extending generally parallel to each other and a second plurality of vertical members **122** extending generally parallel to each other and orthogonal to the first plurality of horizontal members **120**. In an exemplary embodiment, adjacent horizontal members **120** are spaced from each other by a distance of about 4 inches, and adjacent vertical members **122** are spaced from each other by a distance of about 1 inch, although those skilled in the art will recognize that members **120**, **122** can be spaced different distances from those listed above.

(19) While fence **110** in the form of an orthogonal grid is shown in FIG. **1**, those skilled in the art will recognize that fence **110** can be constructed using other configurations, such as a hexagonal configuration commonly found in chicken wire (not shown) or other configurations.

(20) Guard **140** is shown in FIG. **2** as being generally U-shaped in cross section, with a first portion **142** of guard **140** located on a first side of fence **110**, a second portion **144** of guard **140** located on an opposing side of fence **110**, and a connecting portion **146** below fence **110**, connecting first portion **142** with second portion **144**. Guard **140** can extend upwardly from the bottom end **114** of fence **110** a distance of between about 2 inches and about 4 inches, although those skilled in the art will recognize that guard **140** can extend upward a distance less than about 2 inches or greater than about 4 inches.

(21) While guard **140** is shown as extending on both sides of fence **110**, those skilled in the art will recognize that guard **140** can extend only along a single side of fence **110**, and omit second portion **144** and connecting portion **146**. Further, while fence **110** is shown as a mesh fence, guard **140** can be solid so that no openings extend through guard **140**.

(22) In an exemplary embodiment, fence **110** can be constructed from carbon steel that is galvanized or rubber coated to retard rust. Additionally, guard **140** can be constructed from sheet metal, which can be galvanized or rubber coated to retard rust. Alternatively, guard **140** can be constructed from a polymer, such as, for example, PVC, nylon, tetrafluoroethylene, a rubber, or other suitable material that will withstand an exterior environment. Guard **140** is sufficiently thin to allow assembly **100** to be rolled into a coil for shipping and storage.

(23) If both fence **110** and guard **140** are constructed from metal, guard **140** can be welded to fence **110** where fence **110** contacts guard **140**. By way of example only, guard **140** can be welded to fence **110** at welds **150**, **152**, shown in FIG. 2.

(24) If guard **140** is a polymer, guard **140** can be heat applied to bottom end **114** of fence **110** such that first portion **142** of guard **140** is melted to second portion **144** of guard **140** at least around some of horizontal members **120** and/or vertical members **122**, forming a rigid connection between first portion **142** and second portion **144** of guard **140** and locking guard **140** to fence **110**.

(25) While fence **110** is contemplated as being a rigid structure, thereby enabling assembly **100** to be coiled in a roll for shipping and storage, the present invention does not rule out the use of guard **140** with a different type of fence, such as, for example, a cyclone fence.

(26) Referring to FIG. 3, an alternative embodiment of a mesh fence assembly **200** (“assembly **200**”) according to an alternative exemplary embodiment of the present invention is shown. Assembly **200** can include fence **100** as described above. Instead of guard **140** that wraps around both sides of fence **110**, a guard **240** extends only along one side of fence **110** and extends between adjacent horizontal members **120**, with the lowest of the adjacent horizontal members **120** being at the bottom of fence **110** so that guard **240** is proximate to the ground.

(27) If both fence **110** and guard **240** are constructed from metal, guard **240** can be welded to fence **110** at or along horizontal members **120**.

(28) Referring to FIG. 4, an alternative embodiment of a mesh fence assembly **300** (“assembly **300**”) according to an alternative exemplary embodiment of the present invention is shown. Assembly **300** can include fence **100** as described above. Instead of guard **140** that wraps around both sides of fence **110**, a guard **340** extends only from the bottom of fence **110**.

(29) If both fence **110** and guard **340** are constructed from metal, guard **340** can be welded to fence **110** at or along the bottom of fence **110**.

(30) Referring to FIG. 5, a grass mat **400** (“mat **400**”) according to the present invention is shown. Mat **400** can be a continuous horizontal mat that is placed under the centerline of a fence, such as fence **110** discussed above, a fence such as fence **500**, or any other type of fence. Fence **500** is a post and rail fence with posts **502** and a plurality of horizontal rails **504**. Typically, three rows of horizontal rails **504** are used.

(31) Mat **400** can be constructed from a heavy flexible polymer or rubber, including recycled tires. Optionally, mat **400** can be green in color. Mat **400** is used to eliminate the need to spray, trim, or otherwise remove weeds from around a fence. Mat **400** can also be used to discourage pets or other animals from digging under the fence.

(32) Mat **400** can be cut to a desired length, such as the distance between adjacent posts **502** as shown in FIG. 5. Mat **400** can also be provided in differing widths, depending on the amount of grass coverage around fence **500** that is desired. Exemplary widths can be between about 4 inches and about 12 inches, although those skilled in the art will recognize that other widths can be used.

(33) Optionally, as shown in FIGS. 6 and 7, a modified mat **400'** can be used. Mat **400'** is similar to mat **400**, but with a longitudinal slot **402** extending along the centerline of mat **400'**. Slot **402** is sized to allow a horizontal member **120** of a fence, such as fence **110**, to be inserted therein for

stability.

(34) Mat **400** or **400'** can be between about  $\frac{1}{2}$  inch and about  $\frac{3}{4}$  inch thick, although those skilled in the art will recognize that mat **400**, **400'** can be thinner or thicker. Mat **400**, **400'** can have an thickness at edge **406** of between about  $\frac{1}{8}$  inch and about  $\frac{1}{4}$  inch, although those skilled in the art will recognize that edge **406** can be thinner or thicker.

(35) As shown in FIG. 7, mat **400'** can have a cambered cross section **410** such that slot **402** is elevated. Cambered cross section **410** helps keep mat **400'** in place and flat on the ground with the weight of the fence on slot **402**. Cambered cross section **410** can be between about  $\frac{1}{4}$  inch and about  $\frac{1}{2}$  inch high.

(36) Optionally, although not shown, stakes or other securing devices can be driven through mat **400**, **400'** and into the ground to prevent mat **400**, **400'** from blowing away or otherwise moving out of place from under a fence.

(37) It will be further understood that various changes in the details, materials, and arrangements of the parts which have been described and illustrated in order to explain the nature of this invention may be made by those skilled in the art without departing from the scope of the invention as expressed in the following claims.

## Claims

1. A fence with guard comprising: a single fence having a bottom end, the bottom end having a bottom horizontal member and a second horizontal member, each attached to a first side of a plurality of non-horizontal members; and a guard at the bottom end of the single fence and encompassing the plurality of non-horizontal members, wherein the guard comprises a first edge, a first planar sheet extending downward from the first edge, a first curved portion extending from the first planar sheet, distal from the first edge, the curved portion curving in a first direction at an obtuse angle around the first horizontal member and then curves in a second direction at an acute angle vertically upwardly along the plurality of vertical members to a second planar sheet, wherein the second planar sheet is parallel to the first planar sheet and extending upwardly to a second edge.
2. The fence with guard according to claim 1, wherein the fence comprises a plurality of separate horizontal members above the second horizontal member.
3. The fence with guard according to claim 2, wherein the plurality of non-horizontal members comprises a plurality of separate vertical members attached to the bottom and the second horizontal members.
4. The fence with guard according to claim 1, wherein the guard extends below the fence.
5. A fence with integrated guard comprising: a mesh fence having a bottom end, the mesh fence having a plurality of vertical members, a first horizontal member and a second horizontal member, each of the first horizontal member and the second horizontal member extending on a first side of each of the vertical members; and a solid guard extending along the bottom end of the fence and engaging each of the vertical members, wherein the solid guard extends upwardly from the bottom end and covers the first horizontal member and stops below the second horizontal member, wherein a first planar portion of the guard extends vertically downwardly along a second side of the plurality of vertical members, then extends horizontally along a bottom of each of the plurality of vertical members past the first horizontal member, then curves in a first direction at an obtuse angle around the first horizontal member to the first side of the plurality of vertical members, and then curves in a second direction at an acute angle vertically upwardly to a second planar portion, the second planar portion extending along the plurality of vertical members.
6. The fence with integrated guard according to claim 5, wherein the fence has two opposing sides and wherein the guard extends along one of the opposing sides.
7. The fence with integrated guard according to claim 6, wherein the guard extends along a second of the opposing sides.

8. The fence with integrated guard according to claim 5, wherein the guard has a generally U-shaped cross section.
  9. The fence with guard according to claim 5, wherein the guard extends at an oblique angle between the first side of the non-horizontal members and the bottom horizontal member.
  10. The fence with integrated guard according to claim 5, wherein the guard is welded to the fence.
  11. The fence with integrated guard according to claim 5, wherein the fence with integrated guard is configured to be rolled into a coil.
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